

From last time:

Prop (R 1.14) If x, y, z are elements of a field F , then

$$a) \quad x + y = x + z \quad \Rightarrow \quad y = z$$

$$b) \quad x + y = x \quad \Rightarrow \quad y = 0 \quad (0 \text{ is unique})$$

$$c) \quad x + y = 0 \quad \Rightarrow \quad y = -x \quad (\text{negatives are unique})$$

$$d) \quad -(-x) = x$$

Proof

a) Suppose $x + y = x + z$. Let $-x$ be the additive inverse of x , and add to both sides.

$$-x + (x + y) = -x + (x + z).$$

Use associativity:

$$(-x + x) + y = (-x + x) + z.$$

Use inverse axiom:

$$0 + y = 0 + z$$

Use zero axiom:

$$y = z$$

b) $x + y = x$ implies $x + y = x + 0$ by zero axiom.

This shows $y = 0$ by part (a).

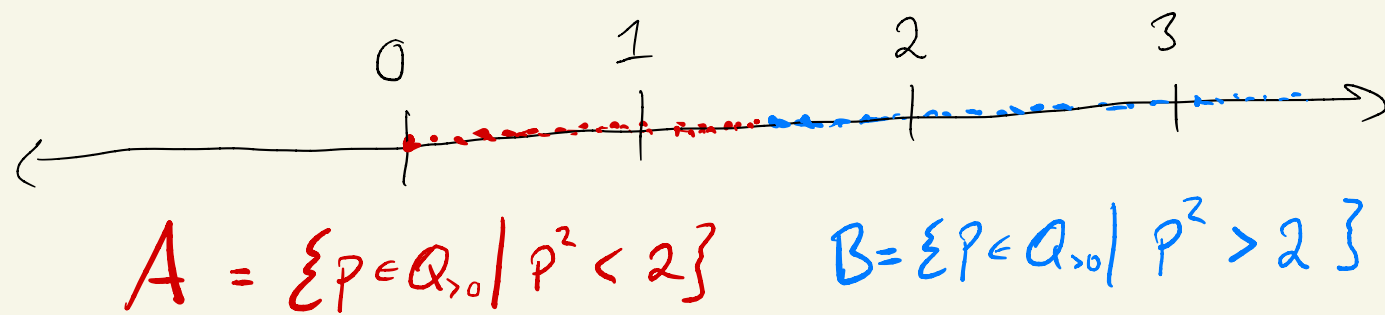
c) $x + y = 0$ implies $x + y = x + x$ by inverse axiom.

Thus, $y = -x$ by part (a).

d) By inverse axiom, $x + (-x) = 0$. By commutativity, $(-x) + x = 0$.

By part (c), $x = -(-x)$.

The rational numbers have "gaps"



What does this mean?

How do we fill them?

Prop A has no maximum and B has no minimum.

Proof: A has no max

Suppose $p \in A$. We must find $\varepsilon > 0$ s.t., $p + \varepsilon \in A$.

We need $(p + \varepsilon)^2 < 2$,

$$\text{equiv. } \Leftrightarrow p^2 + 2p\varepsilon + \varepsilon^2 < 2$$

$$\text{equiv. } \Leftrightarrow \varepsilon(2p + \varepsilon) < 2 - p^2$$

$$\text{equiv. } \Leftrightarrow \varepsilon < \frac{2 - p^2}{2p + \varepsilon}$$

$$\text{implied by } \Leftrightarrow \varepsilon < \frac{2 - p^2}{2p + 1} \text{ and } \varepsilon < 1.$$

$$\text{So } \varepsilon = \min\left(\frac{1}{2} \frac{2 - p^2}{2p + 1}, 1\right) \text{ works.}$$

B has no min

Suppose $p \in B$. We must find $\varepsilon > 0$ s.t., $(p - \varepsilon)^2 > 2$,

$$\Leftrightarrow p^2 - 2p\varepsilon + \varepsilon^2 > 2$$

$$\Leftrightarrow -2p\varepsilon + \varepsilon^2 > 2 - p^2$$

$$\Leftrightarrow \varepsilon(2p - \varepsilon^2) < p^2 - 2$$

$$\Leftrightarrow 2p\varepsilon < p^2 - 2$$

$$\Leftrightarrow \varepsilon < \frac{p^2 - 2}{2p}$$

$$\text{So } \varepsilon = \frac{p^2 - 2}{4p} \text{ works.}$$

Rudin's proof:

Prop. A has no maximum and B has no minimum.

Proof: Let $\phi: \mathbb{Q}_{>0} \rightarrow \mathbb{Q}_{>0}$ be the function $\phi(p) = p - \frac{p^2 - 2}{p + 2}$.

ϕ has two special properties

$$1) \quad \begin{aligned} p^2 < 2 &\Rightarrow \phi(p) > p \\ p^2 > 2 &\Rightarrow \phi(p) < p \end{aligned}$$

$$2) \quad \left. \begin{aligned} p^2 < 2 &\Rightarrow \phi(p)^2 < 2 \\ p^2 > 2 &\Rightarrow \phi(p)^2 > 2 \end{aligned} \right\} \text{ so } \phi \text{ preserves both } A \text{ and } B$$

To prove this, compute $\phi(p)^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}$

So any for any $a \in A$ there is a bigger element $\phi(a)$
and for any $b \in B$ there is a smaller element $\phi(b)$.

□

Lemma If $a, b \in \mathbb{Q}_{>0}$ then $a < b \Leftrightarrow a^2 < b^2$.

i.e. $x \mapsto x^2$ is a strictly increasing function $\mathbb{Q}_{>0} \rightarrow \mathbb{Q}_{>0}$.

Proof Suppose $a < b$.

Mult by a : $a^2 < ab$

Mult by b : $ab < b^2$

transitivity: $a^2 < b^2$.

Suppose $a^2 < b^2$.

Equiv, $0 < b^2 - a^2$

$\Leftrightarrow 0 < (b-a)(b+a)$

$\Leftrightarrow 0 < (b-a)$ because $b+a$ positive.

Prop B is the set of upper bounds of A .

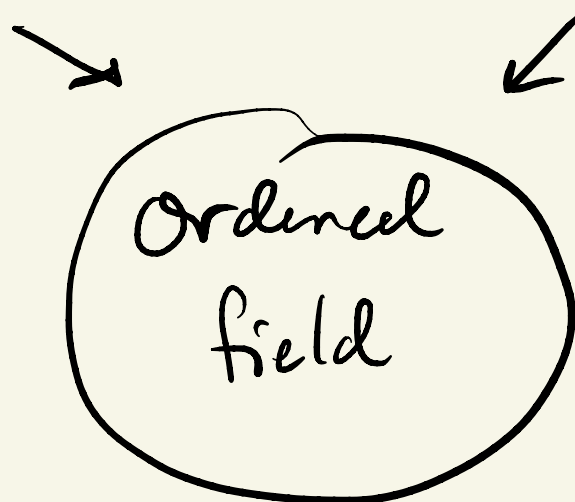
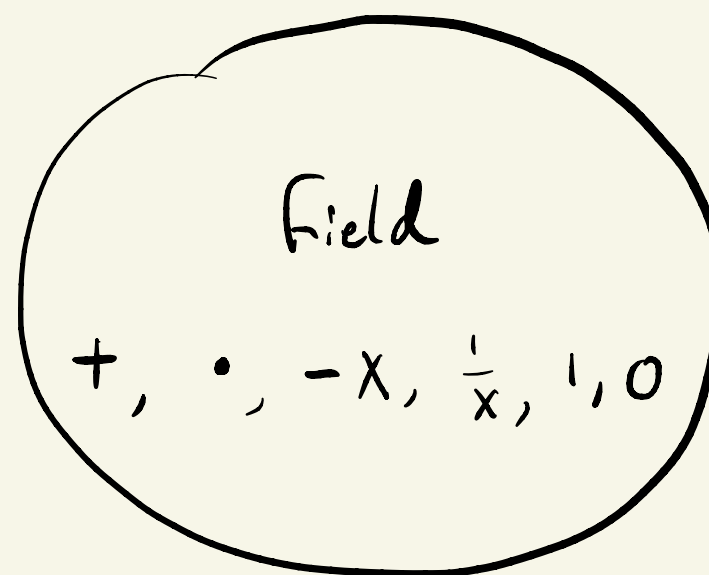
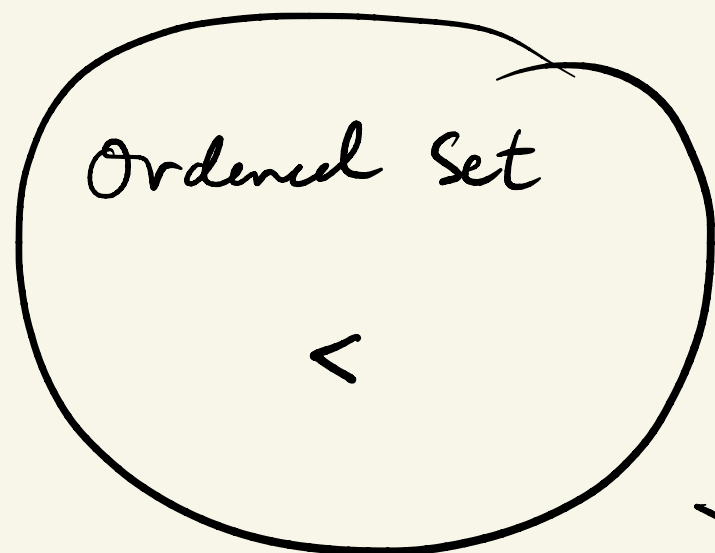
- If $y \in B$ then for all $x \in A$, $x^2 < 2 < y^2$ so $x < y$,
So y is an upper bound of A .

- If $y \in \mathbb{Q}_{>0} \setminus B$ then $y \in A$, So y not upper bound,
as A has no max.

$$x \in A,$$

Prop $A = \{p \in \mathbb{Q}_{>0} \mid p^2 < 2\}$ has no supremum in $\mathbb{Q}_{>0}$, thus
 $\mathbb{Q}$ does not have the LUBP.

proof: Since B has no minimum,
 A has no least upper bound.



Def An ordered field is a field F with an order $<$ such that

$$i) \quad x < y \Rightarrow x + z < y + z \quad \forall x, y, z \in F$$

$$ii) \quad x > 0 \text{ and } y > 0 \Rightarrow xy > 0 \quad \forall x, y \in F$$

Prop The following are true in any ordered field:

$$a) \quad x > 0 \Rightarrow -x < 0$$

$$b) \quad x < y \Rightarrow zx < zy \quad \text{when } z > 0$$

$$c) \quad x < y \Rightarrow zx > zy \quad \text{when } z < 0$$

$$d) \quad x^2 > 0 \quad \text{for all } x \in F \setminus \{0\}$$

$$e) \quad 0 < x < y \Rightarrow 0 < \frac{1}{y} < \frac{1}{x}$$

proof of a)

$x > 0$ implies $x + (-x) > 0 + (-x)$ by ordered field axiom.

By the axiom of negatives, and axiom of zero, this implies $0 > -x$.

proof of b)

$$x < y$$

$\Rightarrow 0 < y - x$ by adding $-x$, using axiom (i)

$\Rightarrow 0 < z(y - x)$ by axiom (ii)

$\Rightarrow 0 < zy - zx$ by distributivity

$\Rightarrow zx < zy$ by (i)

Theorem (Characterization of $\mathbb{R}$)

- There exists an ordered field $\mathbb{R}$ with the least upper bound property.
- $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield.
- If $\mathbb{R}'$ is another such field, there is a unique bijection $\mathbb{R} \rightarrow \mathbb{R}'$ preserving order, multiplication, and addition.

Constructions

- Dedekind: Dedekind cuts, 1872
- Cantor: Cauchy sequences, 1872

Theorem (Rudin 1.20a, Archimedean Property of $\mathbb{R}$)

For all $x, y \in \mathbb{R}$ with $x > 0$, there exists $n \in \mathbb{N}$ such that $nx > y$.

Def (For culture) Let $x, y \in F$ be two positive elements of an ordered field. If $nx < y$

for all $n \in \mathbb{N}$ then x is infinitesimal w.r.t. y .

Ex Real Laurent series $\sum_{n=-M}^{\infty} a_n X^n$ comprise a non-Archimedean ordered field $\mathbb{R}((X))$.

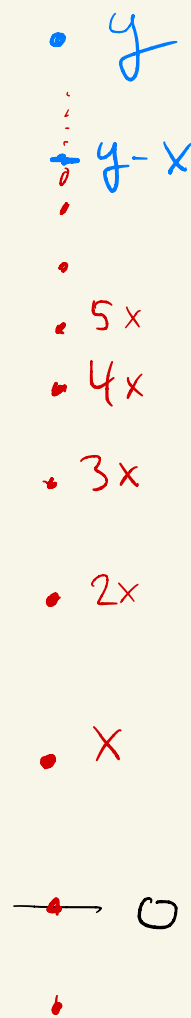
Proof: Let $A = \{nx \mid n \in \mathbb{Z}\} \subset \mathbb{R}$.

Suppose for contradiction that y is an upper bound of A .

Then, A has a least upper bound $\alpha = \sup A$.

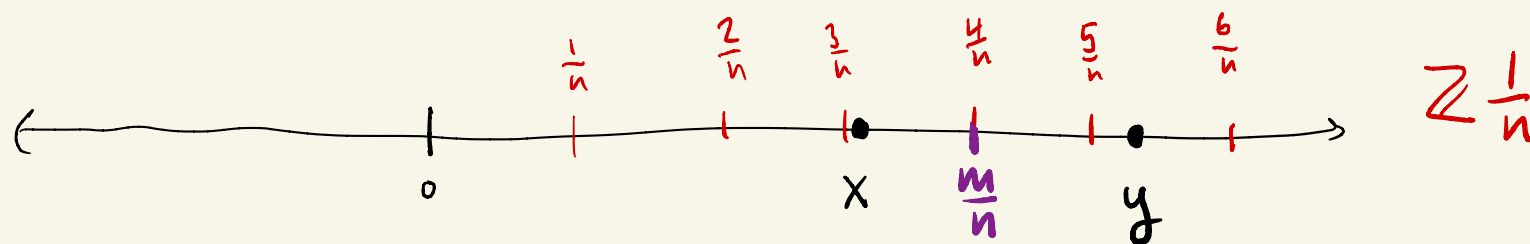
So $\alpha - x < mx$ for some $m \in \mathbb{N}$, because $\alpha - x$ is not an upper bound.

So $\alpha < (m+1)x$, meaning α is not an upper bound. Contradiction.



Theorem (Rudin 1.20b, Density of $\mathbb{Q}$ in $\mathbb{R}$) For every $x, y \in \mathbb{R}$, with $x < y$,

there exists $p \in \mathbb{Q}$ such that $x < p < y$.



First, the case $0 < x < y$.

There exists n with $n(y-x) > 1$ by Arch. Prop. of $\mathbb{R}$.

Let m be the least element of $\mathbb{N}$ s.t. $m > nx$, so $m-1 \leq nx$.

We have $m \leq nx+1$ and $ny > nx+1$ which imply $m < ny$.

Together, $x < \frac{m}{n} < y$.

If $x \leq 0$, there is $b \in \mathbb{N}$ s.t. $b \cdot 1 > -x$ by arch. prop. of $\mathbb{R}$.

There is $p \in \mathbb{Q}$ s.t. $x+b < p < y+b$, so $x < p-b < y$.

Square roots in $\mathbb{R}$

Thm For any real $y > 0$, there exists a unique $x > 0$ s.t. $x^2 = y$.

proof:

Uniqueness: suppose $x_1^2 = y$ and $x_2^2 = y$.

$$x_1 < x_2 \Rightarrow x_1^2 < x_2^2, \text{ contradiction.}$$

$$x_2 > x_1 \Rightarrow x_2^2 > x_1^2, \text{ contradiction.}$$

We must have $x_1 = x_2$.

Existence: Let $A = \{t \in \mathbb{R}_{>0} \mid t^2 < y\}$. A is

- non-empty: For $y \geq 1$, $\frac{1}{2} \in A$.

For $y < 1$, we have $y^2 < y$, so $y \in A$.

- bounded above: $(y+1)^2 = y^2 + 2y + 1 > y$ so $(y+1)^2 > t^2 \forall t \in A$.

Thus, $y+1$ is upper bound of A .

Let $x = \sup A$.

Suppose $x^2 < y$. Then $(x + \varepsilon)^2 < y$ for $\varepsilon = \frac{1}{2} \min\left(1, \frac{y - x^2}{2x+1}\right)$

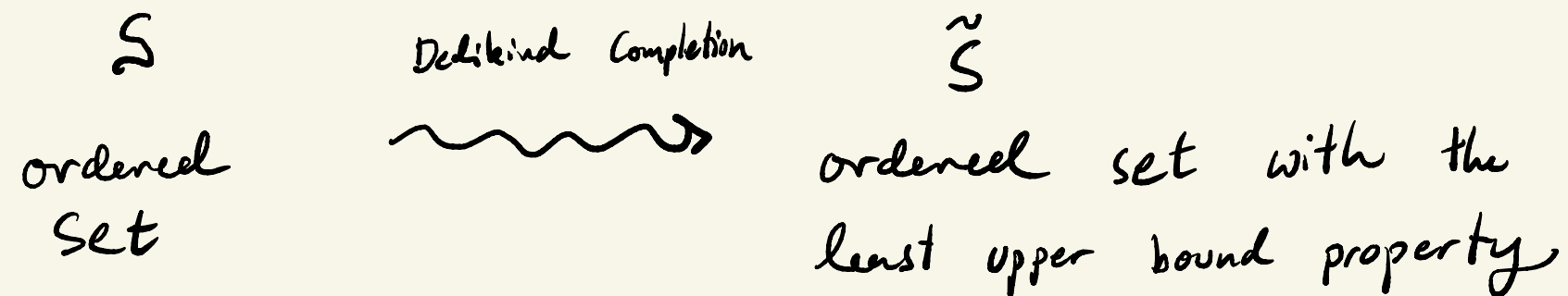
So x is not an upper bound of A .

Suppose $x^2 > y$. Then $(x - \varepsilon)^2 > y$ for $\varepsilon = \frac{1}{2} \min\left(x, \frac{x^2 - y}{2x}\right)$

So x is not the least upper bound.

This leaves only the possibility $x^2 = y$.

Dedekind Completion (For culture)



Definition for $S = \mathbb{Q}$

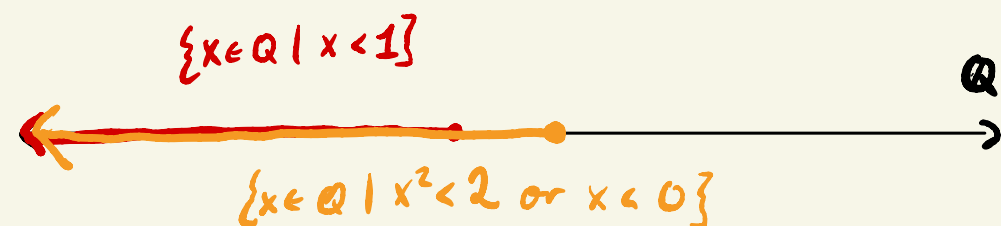
Def A cut is a subset $\alpha \subset \mathbb{Q}$ s.t.,

I) α is not empty

II) If $p \in \alpha$, $q \in \mathbb{Q}$, and $q < p$ then $q \in \alpha$

III) If $p \in \alpha$ then $p < r$ for some $r \in \alpha$.

Def As an ordered set, $\mathbb{R}$ is the set of cuts of $\mathbb{Q}$, with $\alpha < \beta$ if $\alpha \subsetneq \beta$.



Thm (Rudin Ch1 Appendix)

- The Dedekind completion of $\mathbb{Q}$ has the least upper bound property
- We can extend $+$ and $\cdot$ from $\mathbb{Q}$ to $\mathbb{R}$, making it an ordered field.